

Spatio-temporal propagation and reconstruction for low-light video enhancement

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ABSTRACT

Different from the single low-light image enhancement, the key for video enhancement is to improve temporal stability in spatial and temporal varying manner. However, most methods fail to effectively align images or features of adjacent frames to fuse spatio-temporal information and achieve limited enhancement performance, especially in dynamic videos with scene changes and moving targets. In this paper, we propose a spatio-temporal propagation and reconstruction (STPR) network to enable video representation learning for low-light video enhancement. Specifically, we develop a recursively supervised-based pyramid residual dense structure for extracting expressive context features from video sequences, the unique structural design enlarges the receptive field in multi-scale space with limited training parameters. Moreover, to strengthen the interaction of across-frame features, a feature propagation subnet is present to achieve precise alignment and effective fusion of spatio-temporal features. After obtaining the aggregated features of adjacent frames, the spatio-temporal feature reconstruction subnet exploits temporal dependence to improve the quality of target frames and enforce temporal consistency. Experimental results on several low-light video datasets demonstrate that our network outperforms state-of-the-art methods both in quantitative and visual comparisons.

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1. Introduction

Images or videos captured under poor illumination conditions often severely affect the performance of downstream vision tasks, such as recognition, tracking, and classification. The tasks of low-light image/video enhancement are to reconstruct the corresponding visually pleasing counterpart from a given low-quality image/video. Recently, with the rapid development of deep learning in the field of image processing, low-light image enhancement methods have achieved significant advances. However, severe artifacts and flickering occur when low-light image enhancement methods are applied directly to low-light video enhancement. Since the existing image enhancement methods only utilize spatial information of the target frame and ignore the temporal dependence of the video, which results in temporal inconsistency of the enhancement results. In addition, the video output format for most common devices is sRGB, so there is an urgent need for video enhancement algorithm in the sRGB format to improve the quality of low-light videos.

At present, tentative efforts have been made in applying computer vision techniques to improve the quality of low-light videos. These methods can be broadly classified into two paradigms: 1) Frame-by-frame video enhancement methods [1–4] continue the idea of single low-light image enhancement methods, which expand the perceptual field for extracting texture information by stacking filters or convolutional layers to reduce noise and artifacts in low-light videos; 2) Considering the continuous property of the video, multi-frame video enhancement methods [5–9] explore temporal dependence to ensure temporal consistency and improve the enhancement effect.

Existing frame-by-frame low-light video enhancement methods extract spatial features from the target frames for recovering the quality of the entire video, i.e., frame-level quality enhancement. For example, learning-based methods [2,3] extract spatial features by carefully designing the siamese network, and employing self-consistency loss on the output to encourage temporal stability. However, these methods are designed for static videos, and the recovery performance would be limited on dynamic videos. One of the representative works is SMID [2] failed to achieve unsatisfactory enhancement results as shown in Fig. 1. To overcome this problem, Zhang et al. [4] proposed synthesizing video sequences by learning motion fields (optical flow) from the single

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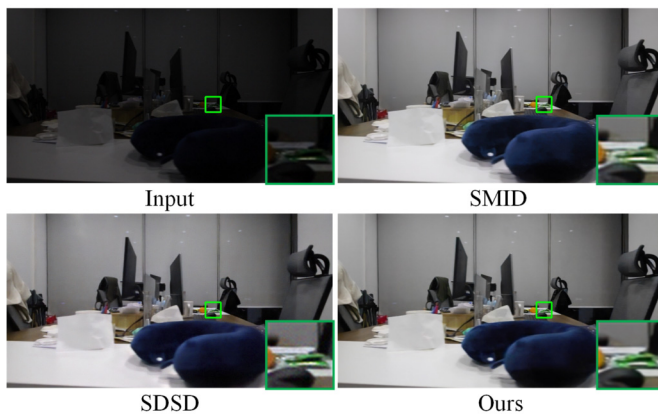


Fig. 1. A comparison of our method with the SOTA method SMID [2] and SDSD [9]. Our result exhibits clearer details with distinct contrast and less noise.

image, but the randomly generated motion fields cannot accurately simulate real low-light video and introducing uncertainty to video enhancement model. These works enhance the target frames only in a spatial-varying manner, thus ignoring the temporal properties of the video sequence. To make full use of multi-frame information, image sequences are adopted as input for obtaining spatio-temporal features. 3D convolutions [6,8,7] are employed to explore features of the successive frames. These methods omit the image or feature alignment module, and a large number of network parameters are required to increase the perceptual field for information extraction. Later, Wang et al. [9] proposed a progressive alignment method SDSD to fuse the adjacent frame information for target frame enhancement, and this implicit alignment failed to restore the desired enhancement effect as shown in Fig. 1. Although considerable progress have been achieved, there are still several problems with low-light video enhancement.

Most works perform only frame-by-frame video enhancement and ignore temporal information of video sequence. Temporal stability cannot be ensured by simply imposing a strong self-consistency on the enhancement results. Naturally, multi-frame spatio-temporal information is considered for improving temporal consistency. However, spatial and temporal information have different characteristics for enhancement. Specifically, in dynamic videos with scene changes and moving targets, spatial information is more suitable for recovering fast-moving regions than temporal information, while temporal information is more effective for enhancing the quality of slow-moving regions. Existing methods have many difficulties in extracting the spatio-temporal information of low-light videos with target motion and global motion. Therefore, the effective exploration of expressive spatio-temporal information for video quality improvement is a problem worth investigating.

Moreover, multi-frame enhancement methods mostly perform low-light video enhancement using spatio-temporal information through the fusion module without frame-level alignment. Practically, the method without alignment cannot effectively fuse multi-frame information by only fixed concatenation, summation, and convolution compared to the method with alignment, which weakens the video enhancement performance, and the feature-wise alignment yields a greater performance gain than image-wise alignment [10]. However, alignment methods are highly dependent on the accuracy of motion field estimation. Commonly used methods for moving field estimation include optical flow and deformable convolution, where optical flow alignment is less robust to occlusion, and deformable convolution alignment is characterized by offset diversity and learning instability [11]. Therefore, to efficiently aggregate and propagate spatio-temporal features of adjacent frames, it is necessary to perform accurate frame-level align-

ment operations in combination with the advantages of the motion field approach.

To address the above limitations, we propose a spatio-temporal propagation and reconstruction (STPR) network to enable video representation learning for low-light video enhancement. The key insight is to exploit the temporal dependence of the video sequences to enforce temporal consistency. Such a pipeline has the following key merits. Firstly, a recursively supervised-based pyramid residual dense structure extracts expressive context features in multi-scale space with limited learning parameters, so that a lightweight model can be constructed. Secondly, a feature propagation subnet is present to achieve accurate alignment and effective fusion of deep spatio-temporal features, which can strengthen the interaction of across-frame information. Besides, to further facilitate the representation of spatial and temporal features in the videos, the spatio-temporal feature reconstruction subnet is constructed to recover the target frame with aligned and aggregated features. The main contributions are summarized as follows:

- We propose a novel STPR network that explores temporal dependence to enhance temporal stable low-light video adaptively.
- A recursively supervised pyramid residual dense structure is developed for expressive context feature extraction in multi-scale space, which enlarges the receptive field with limited learning parameters.
- We present a feature propagation subnet to precisely align and fuse the features of adjacent frames, which is beneficial for the efficient exploitation of spatio-temporal information.
- Extensive experiment results of quantitative and visual comparison demonstrate that our method outperforms the SOTA method and effectively mitigates video flickering.

2. Related work

2.1. Low-light image enhancement

The low-light video enhancement task is similar to the low-light image enhancement task, one may employ low-light image enhancement methods to enhance low-light video in a frame-level manner. Conventional image enhancement methods, such as histogram equalization [12] and gamma correction [13], enhance the quality of images by boosting contrast and extending dynamic range. In addition, single-scale [14] and multi-scale [15] methods based on Retinex theory have been proposed for reconstructing color and illumination. In order to recover the suitable illumination and noise-free reflection components of the image, some improved Retinex methods [16–18] incorporate complex priors. For example, Guo et al. [18] proposed to impose a structure prior to refining the initial illumination map for improving the quality of entire images.

In recent years, low-light image enhancement methods based on deep learning have become a hot research topic [19–21]. Lore et al. [22] firstly proposed a deep autoencoder-based method (LLNet) to identify features and adaptively enhance low-light images. Lv et al. [6] proposed a multi-branch network called MBLEN for extracting and fusing features from different layers. Lately, methods combining Retinex theory and deep learning have been developed [23–25], and DeepUPE [26] estimated an image-to-illumination mapping to enhance underexposed images. Considering the variability of different regions in low-light images, Xu et al. [27] implemented low-light image enhancement by collectively exploiting the Signal-to-Noise-Ratio-aware transformer and convolutional model. Meanwhile, Yang et al. [28] proposed a semi-supervised learning method to recover the linear band representation of the low-light images. Also, many unsupervised learning methods

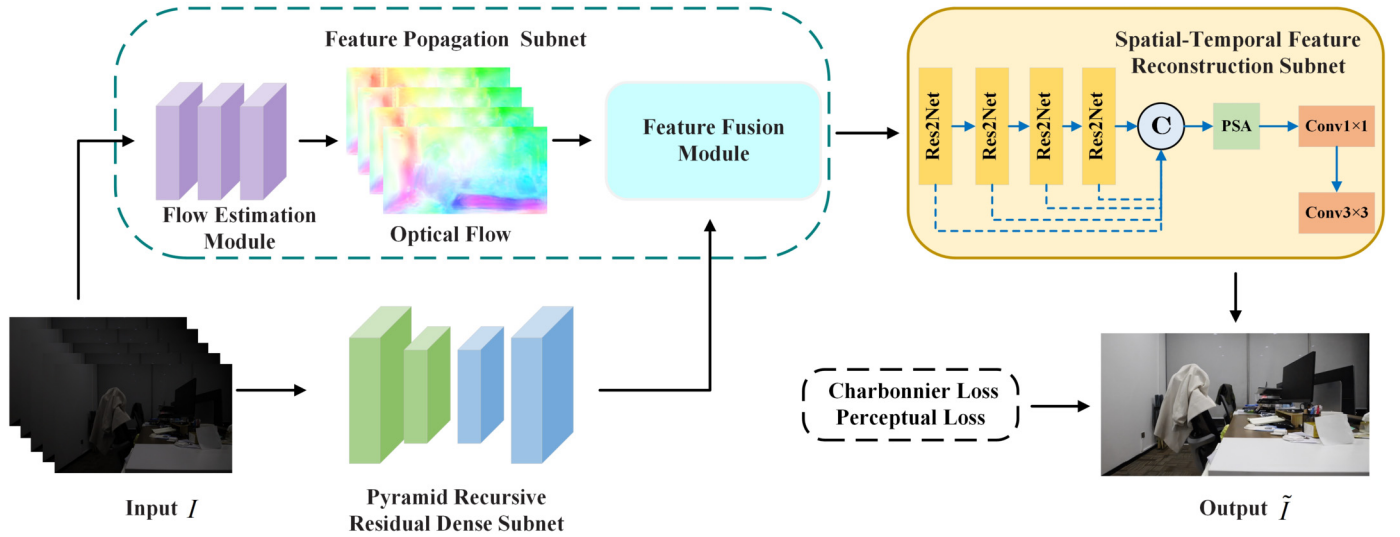


Fig. 2. Framework of the proposed spatio-temporal propagation and reconstruction (STPR) network. Expressive context features from the adjacent frames are extracted by pyramid recursive residual dense subnet, the feature propagation subnet consists of flow estimation module and feature fusion module that aligns and fuses the multi-frame information, then the spatio-temporal feature reconstruction subnet exploits temporal dependence to enhance the target frames.

[29,30] have been presented for image enhancement. For instance, Li et al. [30] exploited the network to learn deep curve estimation for adjusting pixel dynamic range and achieved impressive results. However, the above methods are designed for enhancing a single low-light image, which cannot utilize the temporal information of the adjacent frames and are not suitable for low-light video enhancement tasks.

2.2. Low-light video enhancement

Low-light video enhancement is still a challenging topic due to the difficulty of capturing low-light video datasets and the high hardware processing requirements. A general solution is to continue the single image enhancement method with a frame-to-frame approach for video enhancement. Zhang et al. [1] presented an underexposed video enhancement method using perception-driven progressive fusion. To ensure the temporal consistency of the enhanced video, not only spatial features but also self-consistency loss encourage temporal stability were adopted in SMID [2] and SIDGAN [3]. These works were designed based on static video datasets. In order to extend to dynamic videos, Zhang et al. [4] proposed to learn and infer the motion field from a single image, but the video sequences synthesized from randomly generated optical flow are different from the real captured video, which introduces uncertainty to the enhancement model. To make better use of the multi-frame information of the video. Wang et al. [5] proposed to enhance the low-light video using an LSTM-based neural network but requires several videos taken with the camera beforehand for noise model calibration. 3D convolution has been employed in MBLIVEN [6] and SMOID [8] to process image sequences and aggregate adjacent frame features for exploiting the spatio-temporal information. These methods have poor performance gains due to not being equipped with feature alignment modules. Later, a residual learning structure called VLN [7] is proposed for lighting videos. Wang et al. [9] proposed a network based on Retinex theory to learn information from multi-frame images, progressively alignment and fusion to achieve noise reduction and illumination enhancement of videos. However, these methods still have more improvement potential in the exploration and exploitation of temporal dependence.

2.3. Alignment and fusion

Alignment and fusion of multi-frame information have been widely used in video recovery tasks such as video super-resolution [31–33], video inpainting [34], and video enhancement [35,36], which proves that effective alignment and fusion are crucial for performance improvement. For motion estimation, FRVSR [37] and BasicVSR [38] aligned frames at the image-level according to the optical flow. In contrast, SDSD [9] used deformable convolution (DCN) [39] to perform motion compensation at the feature-level. Since the sensitivity of optical flow to occlusion, DCN has the characteristics of offset diversity and training instability, BasicVSR++ [11] proposed a second-order flow-guided deformable alignment module, which combines the advantages of optical flow and DCN. Furthermore, in video super-resolution tasks, feature-level alignment yields greater performance gains for video recovery compared to image-level alignment [10]. Motivated by precise alignment and effective fusion, we propose a spatio-temporal propagation and reconstruction network with the capability of effectively exploiting the temporal dependence of videos.

3. Method

In this section, we introduce the core idea of spatio-temporal propagation and reconstruction (STPR) network. As illustrated in Fig. 2, the network consists of pyramid recursive residual dense subnet, feature propagation subnet, and spatio-temporal feature reconstruction subnet, which firstly extracts spatial features, then fuses and propagates to make full use of spatio-temporal information for recovering the target frames.

3.1. Pyramid recursive residual dense subnet

For noise and artifacts reduction, larger perceptual field and multi-scale information provide more context information for better enhancement, especially for videos captured in low illumination environments and scenes with global movement. The spatial feature extract subnet focuses on enlarging the receptive field in multi-scale space with limited learning parameters. Motivated by the design of the recursive residual dense block in [40], we propose a more lightweight and efficient pyramid recursive residual

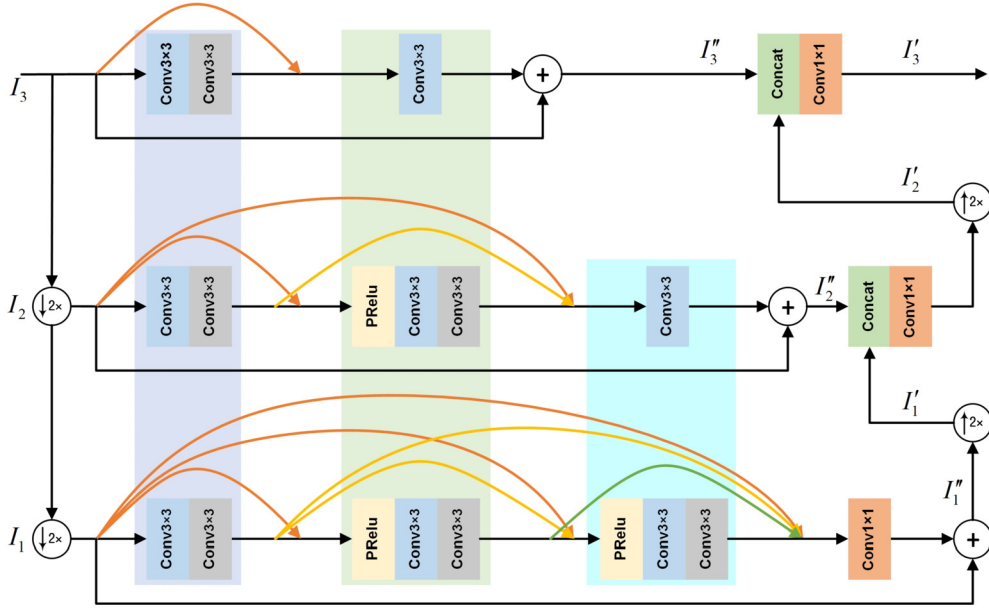


Fig. 3. Architecture of the pyramid recursive residual dense subnet. Video sequences are first fed into a multi-layer pyramid for downsampling. Expressive spatial features are extracted parallelly, and finally, multi-scale spatial features are aggregated step by step upward. Among them, recursive supervision allows convolutions of the same depth to share weights to reduce the model size, dilated convolution expands the perceptual field, and dense residual concatenation helps to retain beneficial spatial features.

dense subnet. Considering rectifiers clip the negative values of feature maps that are irreversibly lost, many studies have argued against the unlimited use of nonlinear activation functions [41–44]. Therefore, we decrease the number and improve use of activation functions, still ensuring that the output of each convolutional layer attains sufficient nonlinearity through the activation function, and the last pointwise convolutional layer sees non-rectified output of all the previous layers. Since multi-scale spatial features in deep networks are also crucial for enhancement performance, simultaneously to improve computational efficiency and reduce learning parameters, we reduce the 1×1 convolutional layers after the residual dense block (RDB) [45] concatenation and try to do more processing at the higher scale space in our proposed pyramidal subnet. In addition, recursive supervision is also employed to optimize the learning of neural networks.

Specifically, pyramid recursive residual dense subnet is introduced to transform the input frames $I \in \mathbb{R}^{N \times H \times W \times 3}$ to spatial feature $I' \in \mathbb{R}^{N \times H \times W \times 96}$ ($N = 5$ denotes a five frame image sequence, H and W represent the height and width of the input frames, respectively). The detailed structure of the proposed subnet is shown in Fig. 3, each frame is firstly downsampled by averaging pooling and then input to different scale layers. Since recursive supervision can reduce learning parameters and increase the depth of the network layers for enlarging the perceptual field, dilated convolution has a larger perceptual field with the same parameters compared to standard convolution. Based on the RDB, we incorporate it with recursive supervision, dilated convolution, and dense residual connection to achieve more efficient spatial feature extraction, where convolutions of the same depth with the same color share weights in Fig. 3. We would like to highlight that feature extraction at different scale spaces of the pyramid structure is performed simultaneously to improve computational efficiency. Then, to alleviate the checkerboard artifacts, we use bilinear interpolation and a 3×3 convolutional layer instead of deconvolutional layer in the upsampling process. Moreover, apart from the feature fusion in every scale space, multi-scale feature fusion is performed to aggregate the features in higher scale space and the intermediate features in current scale space, which is carried out by con-

catenation and a 1×1 convolutional layer. The aggregation process can be written as follows:

$$I'_{i+1} = H_{1 \times 1, i+1}([I'_i, I''_{i+1}]), i = 1, 2, \quad (1)$$

where $H_{1 \times 1, i+1}(\cdot)$ denotes the function of the $(i + 1)$ -th 1×1 convolutional layer. I'_i represents the features in the high-scale space, and I''_{i+1} represents the intermediate features in the current scale space. I'_{i+1} indicates the output of $(i + 1)$ -th 1×1 convolutional layer.

3.2. Feature propagation subnet

Compared with single low-light image enhancement, video enhancement that ignores temporal information leads to the flickering phenomenon. To ensure the temporal consistency of the enhanced video, the adjacent frames features need to be precisely aligned and effectively fused. Motivated by the effectiveness of the grid propagation scheme in super-resolution works [11,33], we propose a feature propagation subnet to facilitate the interaction of across-frame information and improve feature expressiveness. The spatial features extracted from each frame by the forward subnet are passed through the propagation subnet to perform spatio-temporal feature fusion, and the fusion operation $\mathcal{FPS}$ can be written as:

$$f_{FPS}^i = \mathcal{FPS}(f_{i-2}, f_{i-1}, f_i, f_{i+1}, f_{i+2}), \quad (2)$$

where $f_{i-2}, f_{i-1}, f_i, f_{i+1}, f_{i+2}$ represent the features extracted from adjacent frames by the forward feature extraction subnet, respectively. f_{FPS}^i is the spatio-temporal feature after alignment and fusion.

The proposed feature propagation subnet consists of flow estimation module and feature fusion module, as shown in Fig. 2. The flow estimation module adopts the multi-scale motion estimation framework (SpyNet) proposed in [46] to handle moving targets, thus the motion information between adjacent frames and target frame can be obtained for guiding feature alignment. In order to fully aggregate the compensation feature and reduce the

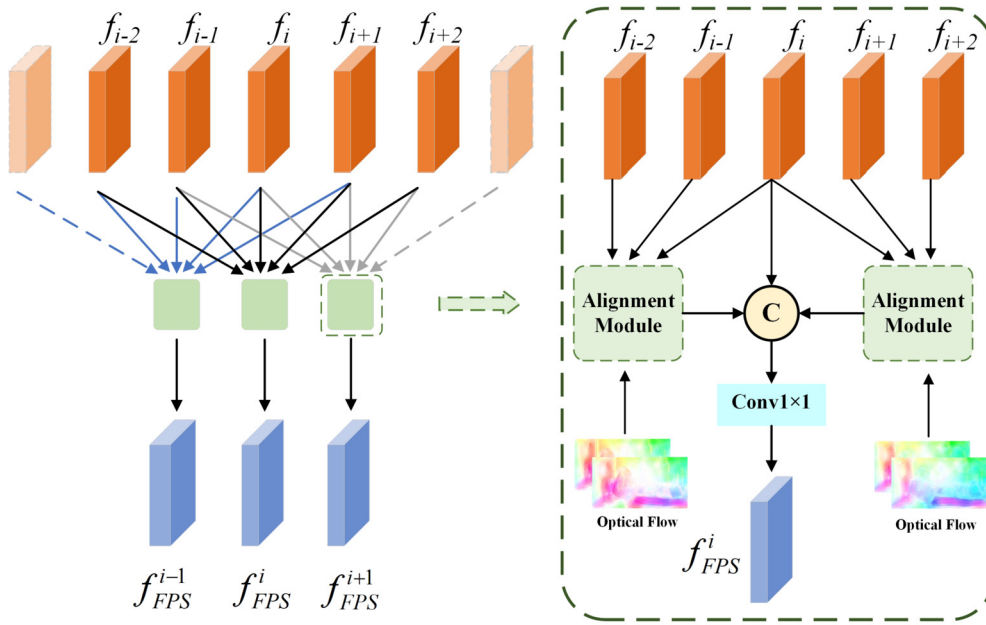


Fig. 4. Architecture of the proposed feature fusion module. Two second-order flow-guided deformable alignment modules are first employed to align features on both sides of the target frames, and an information distillation strategy is introduced at the end of the module to refine aggregated spatio-temporal features and reduce computational complexity.

computational burden, the local fusion module firstly aligns the features on both sides of the neighboring frames by second-order flow-guided deformable alignment, which is shown in Fig. 4. Then, the aligned features are concatenated along the channel dimension to aggregate neighboring frame features. However, since these features contain redundant information and significantly increase the computational complexity. Therefore, we introduce an information distillation strategy to better refine the features. Specifically, a 1×1 convolutional layer is incorporated at the end of the module to ensure the integrity of features with a small number of parameters. The fused features f_{FPS}^i can be expressed as:

$$f_{FPS}^i = \mathcal{R}(\mathcal{C}(\mathcal{A}_{SFGD}(f_{i-2}, f_{i-1}, f_i), f_i, \mathcal{A}_{SFGD}(f_i, f_{i+1}, f_{i+2}))), \quad (3)$$

where $\mathcal{C}$ denotes concatenation operation, and $\mathcal{R}$ denotes 1×1 convolutional layer. Next, we detail the alignment procedure for the left side features, and the right side alignment process is defined similarly. $\mathcal{A}_{SFGD}$ is the second-order flow-guided deformable alignment:

$$\begin{aligned} \hat{f}_l &= \mathcal{A}_{SFGD}(f_{i-2}, f_{i-1}, f_i) = \mathcal{D}(\mathcal{C}(f_{i-1}, f_{i-2}); \\ &\quad \mathcal{C}(o_{i \rightarrow i-1}, o_{i \rightarrow i-2}); \mathcal{C}(m_{i \rightarrow i-1}, m_{i \rightarrow i-2})), \end{aligned} \quad (4)$$

where $\mathcal{D}$ denotes deformable convolution. $o_{i \rightarrow i-1}$, $o_{i \rightarrow i-2}$ represent the offsets from i to $i-1$ and $i-2$, respectively. $m_{i \rightarrow i-1}$, $m_{i \rightarrow i-2}$ represent the modulation masks. Offsets and modulation masks are calculated by concatenating warped features $\tilde{f}$ and flows $s_{i \rightarrow i-p}$ ($p = 1, 2$):

$$\begin{aligned} o_{i \rightarrow i-p} &= s_{i \rightarrow i-p} + \mathcal{C}^o(\mathcal{C}(f_i, \tilde{f}_{i-1}, \tilde{f}_{i-2})), \\ m_{i \rightarrow i-p} &= \sigma(\mathcal{C}^m(\mathcal{C}(f_i, \tilde{f}_{i-1}, \tilde{f}_{i-2}))), \end{aligned} \quad (5)$$

where $\mathcal{C}^{[o, m]}$ represent a stack of convolutions, and σ denotes the sigma function. The fused features f_{FPS}^i will be sent to the spatio-temporal feature reconstruction subnet for recovering the target frames.

3.2.1. Spatio-temporal feature reconstruction subnet

To mitigate artifacts and flickering, the spatio-temporal information of the video needs to be exploited effectively. Therefore, we developed a spatio-temporal feature reconstruction subnet to further explore the temporal dependence. We aim to use a similar single image enhancement strategy to improve the quality of the target frame.

The detailed structure of the spatio-temporal feature reconstruction subnet constructed based on Res2Net [47] and the polarized self-attention (PSA) block [48] is shown in Fig. 2. In particular, to take full advantage of the spatio-temporal information of neighboring frames, the skip connection within the module allows propagation and utilization of features. Additionally, we introduce a PSA mechanism suitable for pixel regression to highlight and suppress features adaptively while maintaining high-resolution information in channel and space. The reconstructed subnet is essential for our entire network, which explores the temporal dependence of successive frames to enforce temporal consistency for recovering the video.

3.3. Loss function

The loss function of our network training consists of two parts. **Charbonnier loss.** Instead of using L1 loss and MSE loss, we use Charbonnier loss [3] to supervise the network training, which can better handle outliers and improve the network performance. Charbonnier loss $\mathcal{L}_c$ is expressed as:

$$\mathcal{L}_c = \sqrt{\|\tilde{I} - \hat{I}\|_2^2 + \epsilon^2}, \quad (6)$$

where $\tilde{I}$, $\hat{I}$ denotes the predicted output and ground-truth, respectively. And the regularization term ϵ is set as 10^{-3} in this paper.

Perceptual loss. To improve the perceptual information fidelity of the target frames, we measure the distance between features based on the pre-trained VGG-19 model. The perceptual loss $\mathcal{L}_{per}$ is written as:

$$\mathcal{L}_{per} = \frac{1}{C_n H_n W_n} \left\| \phi_n(\tilde{I}) - \phi_n(\hat{I}) \right\|_2^2, \quad (7)$$

where $\phi_n(\cdot)$ is the feature map of the n -th convolutional layer in VGG-19 model. C_n , H_n , W_n denote the dimensions of the corresponding feature maps, respectively.

Total loss. The total loss function is:

$$\mathcal{L}_{total} = \mathcal{L}_c + \lambda_{per} \mathcal{L}_{per}, \quad (8)$$

where λ_{per} is the balance coefficient.

4. Experiments

4.1. Datasets and evaluate metrics

Two widely used low-light video datasets are adopted to train and evaluate the STPR network: SDSD [9] and SMID [2]. SDSD is the first low-light video dataset with ground-truth captured in real dynamic scenes, including 70 indoor video sequences and 80 outdoor video sequences. Note that, the SDSD dataset is collected in the case of global camera motion. For SMID, which consists of static videos with corresponding long-exposure ground-truth and dynamic videos without ground truth, we transformed the dataset from RAW to sRGB format for our experiments. Our training and testing sets are set up following SDSD [9] and SMID [2].

The Peak Signal-to-Noise Ratio (PSNR) and Structural Similarity (SSIM) [49] metrics are employed to evaluate the performance of enhancement methods, Absolute Brightness Variance (AB) [6] and Mean Absolute Brightness Difference (MABD) [8] are introduced to assess the temporal stability of models. Generally, larger values of PSNR and SSIM represent better video quality, while smaller values of AB and MABD indicate better temporal consistency.

4.2. Experimental settings

The proposed STPR network is implemented on the Pytorch framework. We randomly crop 128×128 patches for training on NVIDIA RTX 3090 GPU, all these patches are transformed by randomly flipping and rotations of 90° , 180° , 270° . The Cosine Annealing learning scheduler [50] is adopted to train the network, and the initial learning rate decays from 4×10^{-4} to 10^{-7} . The total number of network iterations is 300 K. During the training stage, we load and freeze the weights of the pre-trained motion estimation network [51]. We set the hyperparameter λ_{per} as 10^{-2} in our experiments. The mini-batch is set to 12. The network is trained using Adam [52] optimizer with $\beta_1 = 0.9$, $\beta_2 = 0.99$.

4.3. Enhancement results

To prove the effectiveness of our proposed method, we choose eleven low-light enhancement methods for quantitative and visual comparisons. Among them, DeepUPE [26], Zero-DCE [29], DeepLPF [53], DRBN [54], SNR [27] are the representative low-light image enhancement methods, MBLLEN [6], SMID [2], SMOID [8], StableLLVE [4], SGZSL [55], SDSD [9] are the state-of-the-art low-light video enhancement methods.

4.3.1. Evaluation on SDSD dataset

Table 1 shows the results of the quantitative comparison of the SDSD dataset. The PSNR, SSIM, AB, and MABD of our proposed STPR obtain the best scores. In particular, for PSNR, the proposed method significantly outperforms all baselines (exceeds the second-best method by 1.12 dB), which implies that the proposed STPR has the best quality enhancement performance for low-light video. Since the single image enhancement methods ignore temporal information, DeepUPE, Zero-DCE, ZeepLPF, and DRBN obtain lower PSNR and SSIM scores. The SNR introduces the signal-to-noise ratio prior and thus obtains high scores. In contrast, the

Table 1

Quantitative comparison on SDSD dataset. The best results are in bold.

Methods	PSNR $\uparrow$	SSIM $\uparrow$	AB $\downarrow$	MABD ($\times 10^{-4}$) $\downarrow$
DeepUPE [26]	21.82	0.68	-	-
ZeroDCE [29]	20.06	0.61	-	-
DeepLPF [53]	22.48	0.66	-	-
DRBN [54]	22.31	0.65	-	-
SNR [27]	26.71	0.85	-	-
MBLLVEN [6]	21.79	0.65	-	-
SMID [2]	24.09	0.69	12.4056	0.8592
SMOID [8]	23.45	0.69	4.4679	0.7503
StableLLVE [4]	20.57	0.74	11.9585	1.5472
SGZSL [55]	23.89	0.70	24.0161	1.1415
SDSD [9]	24.92	0.73	4.6370	0.7527
Ours	27.83	0.86	4.3923	0.1247

video enhancement methods exploit the complementary information of the sequences, and the PSNR and SSIM scores of SMID, SMOID, and SDSD can be improvement to some extent. For AB and MABD metrics, compared to the SMID and StableLLVE, which enforce temporal consistency using self-consistency loss, SMOID and SDSD use sliding windows to capture spatio-temporal features and therefore obtain better AB and MABD scores. SGZSL is limited by the accuracy of semantic segmentation and gets higher values of AB and MABD. However, our model can take full advantage of the precisely aligned and effectively fused multi-frame spatio-temporal features, which are more advantageous in temporal dependence exploitation. Therefore, the proposed STPR yields the best AB and MABD scores on the SDSD dataset, which indicates that our method is more stable temporal consistency in videos with scene changes and moving targets.

Fig. 5 and Fig. 6 show the visual comparison of indoor and outdoor scenes, respectively. It is clear to see that the STPR has much quality improvement compared to the other results. Although other comparison methods achieve contrast and main structure enhancement, there are still some shortcomings. The recovery results of Zero-DCE suffer from color distortion and noise. Since simple self-consistency loss is enforced to correlate adjacent frames, SMID produces the result of overexposure and substantial loss of detail. The results recovered by the multi-frame enhancement method SMOID suffer from artifacts and blurring, which may be caused by the absence of frame-level alignment and fusion modules. DeepUPE learns to preserve more details in bilateral space, but their results are still heavily affected by noise due to without the denoising mechanism. Considering the noise, the state-of-the-art single image enhancement method SNR and the multi-frame video enhancement method SDSD both integrate an implicit denoising mechanism in their models, but SNR over-smoothing causes detail loss, SDSD still produces severe noise and grid degradation. Comparatively, the proposed STPR yields the best visual perceptual quality, with suitable illumination, clear detail, less noise, and natural color. The main reason is that the pyramid recursive residual dense subnet extracts expressive context features. Meanwhile, the spatial-temporal feature reconstruction subnet exploits the temporal dependence of video through multi-frame features that are precisely aligned and efficiently aggregated by the feature propagation subnet.

4.3.2. Evaluation on SMID dataset

To demonstrate the generalization of our method, we test the comparison methods on the SMID dataset. In Table 2, the quantitative results show that our method obtains the highest PSNR and SSIM scores (exceeds the second-best method by 2.75 dB and



Fig. 5. Visual comparison of low-light video enhancement on the SDSD indoor scenes. It can be seen that the frames enhanced by the proposed STPR have suitable illumination, less noise, as well as natural color.

Table 2

Quantitative comparison on SMID dataset. The best results are in bold.

Methods	PSNR $\uparrow$	SSIM $\uparrow$	AB $\downarrow$	MABD ($\times 10^{-4}$) $\downarrow$
DeepUPE [26]	23.91	0.69	-	-
ZeroDCE [29]	22.62	0.67	-	-
DeepLPF [53]	24.36	0.69	-	-
DRBN [54]	24.42	0.69	-	-
MBLLVEN [6]	22.67	0.68	-	-
SMID [2]	24.78	0.72	3.4529	0.1956
SMOID [8]	23.64	0.71	2.5808	0.3905
StableLLVE [4]	26.25	0.78	3.3516	0.3258
SDSD [9]	26.03	0.75	3.3170	0.2755
Ours	29.00	0.82	2.9757	0.6777

ues in the AB and MABD metrics. Since the SMID testing dataset is static video, the PSNR and SSIM performance of the single-frame image enhancement method and the video enhancement method are comparable. While StableLLVE and SDSD incorporate semantic information and progressive alignment mechanisms, respectively, resulting in better enhancement performance. Besides, to ensure video temporal stability, the video-based methods use self-consistency loss or multi-frame information to enforce temporal consistency for avoiding flickering. The results in Table 2 show that SMID and SMOID perform better in AB and MABD metrics, but obtain lower PSNR and SSIM values. With a special feature propagation subnet. However, the proposed method learns to exploit temporal dependence to enforce temporal stability and maintains a good balance between detail preservation and noise suppression. Therefore, STPR performs better in quantitative results compared with other methods.

In addition, Fig. 7 provides a visual comparison of the SMID dataset. The recovered video of SMID is underexposed. DeepUPE,

0.04, respectively), and achieves comparable results to the best val-

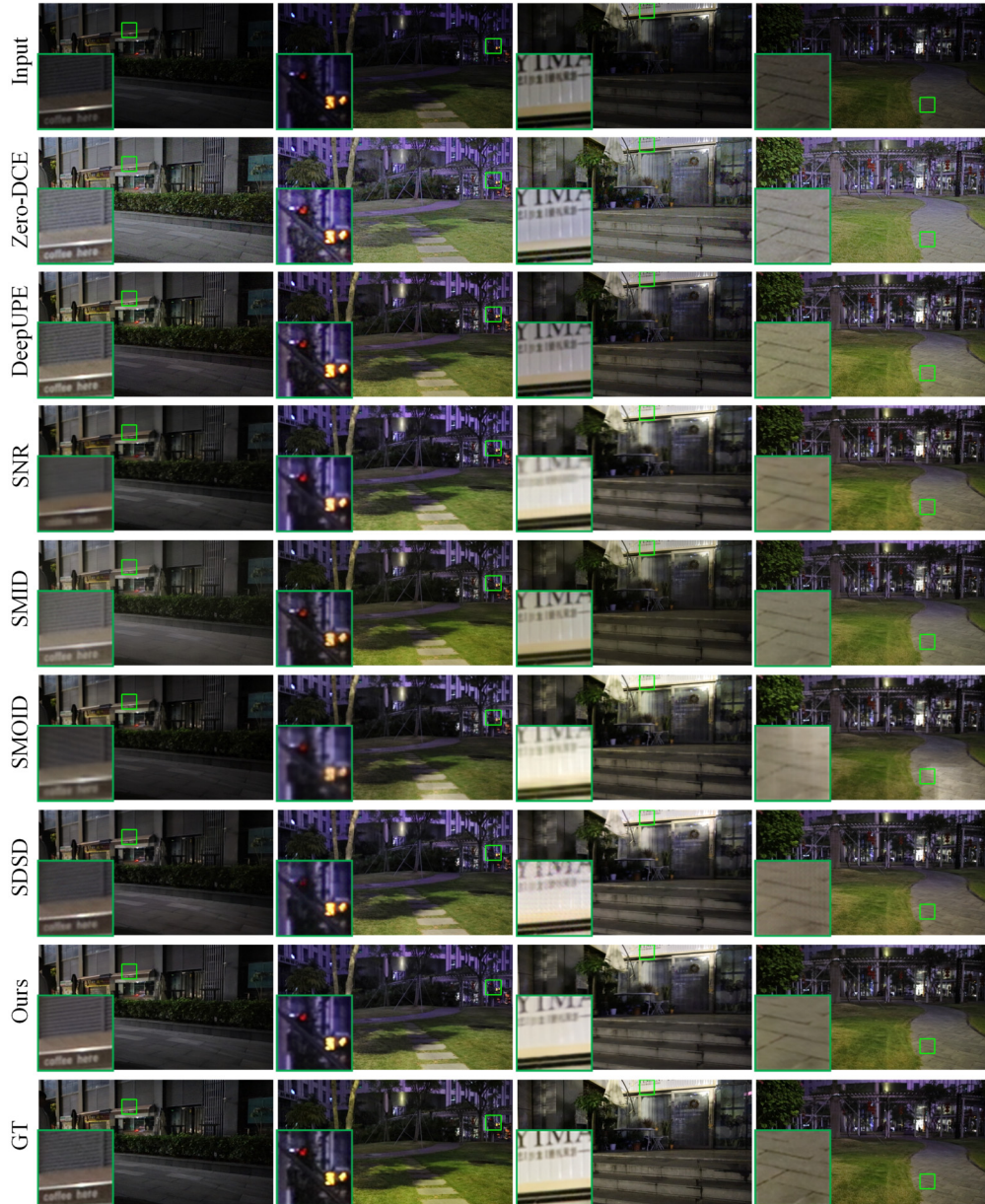


Fig. 6. Visual comparison of low-light video enhancement on the SDSD outdoor scenes. It can be seen that the frames enhanced by the proposed STPR have suitable illumination, less noise, as well as clearer texture details.

Zero-DCE, and SDSD suffer from color distortion and noise. SMOID generates severe artifacts and StableLLVE loses details. Compared with these methods, the enhancement results of the proposed STPR show suitable brightness, less noise, clear details, and natural colors.

4.4. Ablation study

In this part, we perform an ablation study to demonstrate the effectiveness of subnet design and components in STPR, which was conducted on the SDSD indoor and outdoor datasets. We adopt the spatial feature extraction subnet in [40] as Baseline. PRRD represents the pyramid recursive residual dense subnet, FP denotes the feature propagation subnet, and STFR is the spatio-temporal feature reconstruction subnet.

To verify the validity of the PRRD perceptual field and multi-scale design, we investigated the two baselines listed in Table 3. In particular, “Standard-STPR” represents the network that replaces

the dilated convolution of PRRD with standard 2D convolution. The “Single-scale-STPR” is a baseline that retains only the lower branches of the PRRD and does not consider spatial downsampling. The results in Table 3 broadly validate our design choices. It shows that the judicious use of dilation convolution helps the model to learn richer contextual information (by comparing the results in 1st row to the 3rd row). In addition, features at different spatial scales and abstraction levels both help the model recover images with higher signal-to-noise ratios (comparing the result in the 2nd row to that in the 3rd row).

To evaluate the effectiveness of our network components, we start with PRRD and gradually add components. As illustrated in Table 4, the proposed PRRD yields higher PSNR and SSIM values compared to Baseline, which shows that our PRRD can extract expressive context features. The addition of STFR to PRRD provides more PSNR gain than the addition of FP, which is due to the fact that STFR reconstructs the target frame using temporal dependence of video sequences. However, the enhancement performance of the



Fig. 7. Visual comparison of low-light video enhancement on the SMID dataset. It can be seen that the frames enhanced by the proposed STPR have suitable brightness, natural color, as well as clearer texture details.

Table 3

Ablation results of subnet design on SDSD dataset. The best results are in bold.

Methods	SDSD-indoor	SDSD-outdoor
	PSNR/SSIM	PSNR/SSIM
Standard-STPR	27.82/0.88	26.88/0.82
Single-scale-STPR	27.71/0.88	26.58/0.82
STPR	28.22/0.89	27.43/0.82

Table 4

Ablation results of network components on SDSD dataset. The best results are in bold.

Methods	SDSD-indoor	SDSD-outdoor
	PSNR/SSIM	PSNR/SSIM
Baseline	23.39/0.84	21.98/0.77
PRRD	24.40/0.86	24.34/0.79
PRRD+FP	25.03/0.87	25.69/0.81
PRRD+STFR	27.15/0.87	26.64/0.81
STPR	28.22/0.89	27.43/0.82

model without a frame alignment mechanism can be further improved. Thus, we introduce FP to precisely align and effectively fuse spatio-temporal features of adjacent frames. The visual comparison of the ablation experiments in Fig. 8 further shows the effectiveness of our proposed components.

4.5. Model complexity evaluation

In general, better performance gains require models with large-scale parameters. As shown in Table 5, although MBLLEN has the minimum model parameters, PSNR and SSIM scores of the enhancement results perform poorly. In contrast, compared to other low-light video enhancement methods, our model with only 4.14M parameters still achieves significant performance gains, and few learning parameters facilitate us to build a lightweight model and save memory during training.

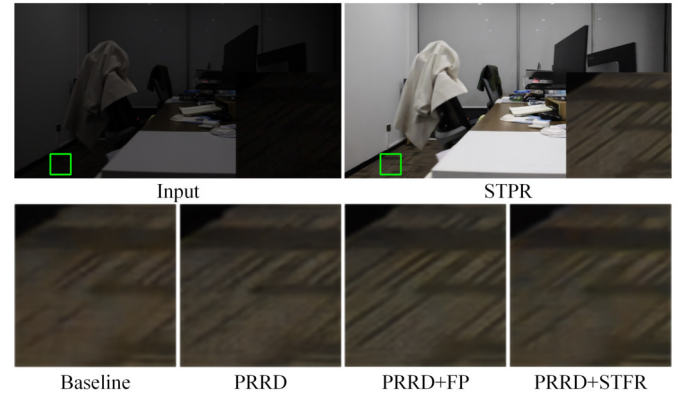


Fig. 8. Visual comparison of the network structure ablation study. PRRD can extract expressive context features. STFR exploits the temporal dependence of video through multi-frame features that are precisely aligned and efficiently aggregated by FP.

Table 5

Quantitative comparison of the network complexity.

Methods	Param(M)↓	PSNR↑
MBLLEN [6]	1.02	21.79
SMID [2]	6.22	24.09
SMOID [8]	3.64	23.45
SGZSL [55]	28.10	23.89
SDSD [9]	4.43	24.92
Ours	4.14	27.83

5. Conclusions

In this paper, we present the STPR network to achieve low-light video enhancement in spatial and temporal varying manner. With the designed pyramid recursive residual dense structure, the spatial feature subnet is capable of enlarging the receptive field in multi-scale space for expressive context feature extraction with limited training parameters. Meanwhile, different from the existing video enhancement methods that explore temporal information.

The adjacent features are precisely aligned and effectively fused by the proposed feature propagation subnet. Furthermore, the spatio-temporal feature reconstruction subnet exploits the temporal dependence of video sequences to enhance the target frame and enforce temporal consistency. Extensive experiments have demonstrated that our network achieves PSNR and SSIM improvement, and the enhanced video benefits from stable temporal consistency. Our network achieves superior performance compared to state-of-the-art methods.

Our feature propagation subnet can efficiently aggregate multi-frame information, which can be extended to fields such as video super-resolution, video denoising, and video action recognition. In addition, the lower parameters of our network reduce memory consumption, but there are still limitations in real-time. In future work, based on STPR, we will combine the fast algorithm [56,57] to optimize the prediction speed deficiency to form a low-complexity and high-efficiency low-light video enhancement method.

CRedit authorship contribution statement

Jing Ye: Conceptualization, Data curation, Methodology, Software, Visualization, Writing – original draft. **Changzhen Qiu:** Writing – review & editing. **Zhiyong Zhang:** Supervision, Writing – review & editing.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

Data will be made available on request.

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